

Repsol Electra 1X

Paraffinic dielectric oil — Inhibited (for the Latin American market)

PRODUCT DATA SHEET

Description

ELECTRA 1X is a light, paraffinic oil designed for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc., specifically for the Latin American market. It is manufactured from highly-refined treated base oils ensuring the absence of solid, polar or water compounds and of corrosive sulphur.

Classification and standards

ELECTRA 1X is an inhibited oil formulated with antioxidant additives, for the Latin American market.

Technical specifications

Values according to ASTM D3487 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		ASTM D1524		Bright and clear	
Colour		ASTM D1500		0.5	0.5
Relative density at 15 °C	g/cm3	ASTM D1298		0.91	0.855
Viscosity at 0 °C	cSt	ASTM D445		76.0	68.3
Viscosity at 40 °C	cSt	ASTM D445		12.0	11.3
Viscosity at 100 °C	cSt	ASTM D445		3.0	2.8
Pour point	°C	ASTM D97		-40	-26
Aniline point	°C	ASTM D611	63		91.8
Interfacial tension	dynes/cm	ASTM D971	40		41
Chemical properties					
Acidity	mg KOH/g	ASTM D974		0.03	<0.01
Corrosive sulphur		ASTM D1275B		Non-corrosive	
Water content	ppm	ASTM D1533		35	29
Inhibitor content	% weight	ASTM D2668		0.3	0.3
Electrical properties					
DDF 60 Hz, 25 °C	%	ASTM D924		0.05	0.002
DDF 60 Hz, 100 °C	%	ASTM D924		0.30	0.167
Breakdown voltage 60 Hz - disk electrodes	kV	ASTM D877	30		47
BDV 60 Hz - VDE 0.04 in (1.02 mm) gap	kV	ASTM D1816	20		28
BDV 60 Hz - VDE 0.08 in (2.03 mm) gap	kV	ASTM D1816	35		

Properties	Units	Method	Min.	Max.	Typical value
BDV impulse, 25 °C, 1-in (25.4 mm) gap	kV	ASTM D3300	145		178
Gassing tendency	uL/min	ASTM D2300		30	
Oxidation stability — Acid-sludge test					
TAN (72 h)	mg KOH/g	ASTM D2440		0.3	0.01
TAN (164 h)	mg KOH/g	ASTM D2440		0.4	<0.01
Sludge (72 h)	% weight	ASTM D2440		0.10	0.01
Sludge (164 h)	% weight	ASTM D2440		0.30	<0.01
Rotating bomb test	min	ASTM D2112	195		255
Health, safety and the environment					
Flash point	°C	ASTM D92	145		186
PCB content	ppm	ASTM D4059		Undetectable	

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.